Artificial Neural Networks (ANN) - Simple Summary

1 Introduction

Artificial Neural Networks (ANN) are models inspired by the human brain. They are used in Machine Learning to learn patterns from data and make predictions.

Idea: Instead of writing rules manually, the model learns from examples.

2 Basic Structure of ANN

An ANN is made of:

- **Input Layer:** Receives data
- **Hidden Layer(s):** Processes data
- **Output Layer:** Gives final result

Example: If we want to predict if an email is spam:

- Input: words in the email
- Output: spam or not spam

3 Neuron (Perceptron)

A neuron takes inputs, multiplies them by weights, adds bias, then applies a function.

3.1 Formula

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

$$y = f(z)$$

Where:

- x_i = inputs
- w_i = weights
- b = bias
- f = activation function

Simple Explanation: Weights = importance of each input Bias = extra adjustment Activation = decision function

4 Activation Functions

They decide the output of a neuron.

4.1 Common Types

1. Step Function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

2. Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$

Output between 0 and 1.

3. ReLU

$$f(x) = \max(0, x)$$

Rule:

- Use ReLU in hidden layers (fast and simple)
- Use Sigmoid for probability outputs

5 Forward Propagation

This is how data moves in the network.

Steps:

1. Input enters the network
2. Multiply by weights
3. Add bias
4. Apply activation function
5. Pass to next layer

Example:

- Input = (2, 3)
- Weights = (0.5, 0.2)
- Bias = 1

$$z = (2 \times 0.5) + (3 \times 0.2) + 1 = 2.6$$

Then apply activation.

6 Loss Function

Measures how wrong the prediction is.

Example (Mean Squared Error):

$$Loss = \frac{1}{n} \sum (y_{\text{true}} - y_{\text{pred}})^2$$

Simple Idea: Small loss = good model Big loss = bad model

7 Backpropagation

This is how the network learns.

Steps:

1. Calculate error (loss)
2. Send error backward
3. Update weights

Rule:

- Adjust weights to reduce error

8 Gradient Descent

Used to update weights.

$$w = w - \eta \frac{\partial Loss}{\partial w}$$

Where:

- η = learning rate

Simple Explanation:

- Move weights step by step toward better values

9 Learning Rate

Controls how fast the model learns.

- Too large → unstable
- Too small → slow learning

10 Overfitting vs Underfitting

Overfitting:

- Model memorizes data
- Performs badly on new data

Underfitting:

- Model is too simple
- Cannot learn patterns

11 Example of ANN Usage

- Image recognition
- Speech recognition
- Recommendation systems (like your restaurant app idea!)

12 Final Summary

- ANN learns from data like the human brain
- Uses neurons, weights, and activation functions
- Forward propagation makes predictions
- Backpropagation improves the model